PROBLEM

ARTIFICIAL INTELLIGENCE – CSA1717

ASSESSMENT TOOL 3 – DRY RUN TEST

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Course Code: CSA1717
Course Name: Artificial Intelligence
Assessment Title: Dry Run Test

QUESTION 1: DRY RUN OF A* SEARCH ALGORITHM

Problem Statement

Consider the following graph with nodes **A, B, C, D, E, and G**. Perform a dry run of the *A Search Algorithm** to find the optimal path from **A** to **G**.

Edge Costs

Edge	**Cost**
A → B	2
A → C	4
B → C	3
B → D	7
B → E	2
C → E	3
D → E	2
D → G	2

Heuristic Values

Node	**h(n)**
A	7
B	6

C	4
D	3
E	2
G	0

Tasks

Perform the A* Search dry run from **A to G**.

For each iteration, identify:

- Current Node
- Open List
- Closed List
- g(n)
- h(n)
- f(n) = g(n) + h(n)

Finally determine:

1. The optimal path.
2. The total path cost.

QUESTION 2: DRY RUN OF MINIMAX WITH ALPHA-BETA PRUNING

Problem Statement

Consider the following game tree. The root node is a **MAX** node and its child nodes are **MIN** nodes. The leaf nodes contain the utility values.

```
      MAX
     /   \
   MIN     MIN
  / | \   / | \
 3  5  6  9  1  2
```

The tree must be evaluated **from left to right**.

Initial Values

Alpha (α) = −∞

Beta (β) = +∞

Tasks

Perform a dry run of the **Minimax Algorithm with Alpha-Beta Pruning**.

For each step, identify:

- Current Node
- Alpha (α)
- Beta (β)
- Selected Value
- Any Pruned Node

Finally determine:

1. The best move for MAX.
2. The final minimax value.
3. The list of pruned nodes.